



Validation Report v3 — one platform, every device

Follows v2. This cycle: the server resize was verified, the legacy sidebar design retired so every page wears the NEXUS chrome and the same bottom navigation, the blue Quant Edge mark made the only logo, the platform validated on iPhone and iPad against a new SR 11-7-style interface standard (UX 11-7), and the #1 quant recommendation tested walk-forward and shipped. Prepared September 24, 2026.

Executive summary

Area	Result	Status
Server resize (512 MB -> 2 GB)	Free RAM 8 -> 1024 MB; swap 508 -> 0 MB; health check 6.4 s -> 12 ms	Validated
One platform chrome	Sidebar layout retired; 12 pages + research now share the terminal topbar and bottom dock	Shipped
One logo	Blue Quant Edge Q redrawn as vector: header, favicon, iOS/Android icons; 5 competing marks retired	Shipped
iPhone fit	Pages zoomed out to 582-1,572 px on a 393 px phone -> all fit; dock reachable everywhere	Fixed
iPhone legibility	Board text under 11 px: 778 of 987 -> 0 (responsive type floor on touch)	Fixed
UX 11-7 scorecard (iPhone)	107/121 criteria pass across 11 screens; research 11/11	Validated
Stop floor (1.25x ATR)	Out-of-sample -0.225R -> -0.115R per idea (n=332)	Shipped
Slow shared reads	market-pulse 21 s -> 7 ms, extended-hours 8 s -> 6 ms (stale-while-revalidate + warm)	Fixed
Data-heavy screens still slow (U8)	LCP 2.7-15 s; one vCPU at load 2.28, scanners take ~64% CPU	Operator

Part A — Revalidating accuracy

A1. The stop-floor intervention, tested before shipping

v2 found stop width to be the strongest loss driver (IC +0.30, holding within every producer). Measured this cycle: the median published stop sat at **0.59x** the stock's own ATR(14) — inside a normal day's range. Two interventions were replayed on the real 5-minute path of every idea (664 ideas), the floor chosen on the first half and scored only on the second:

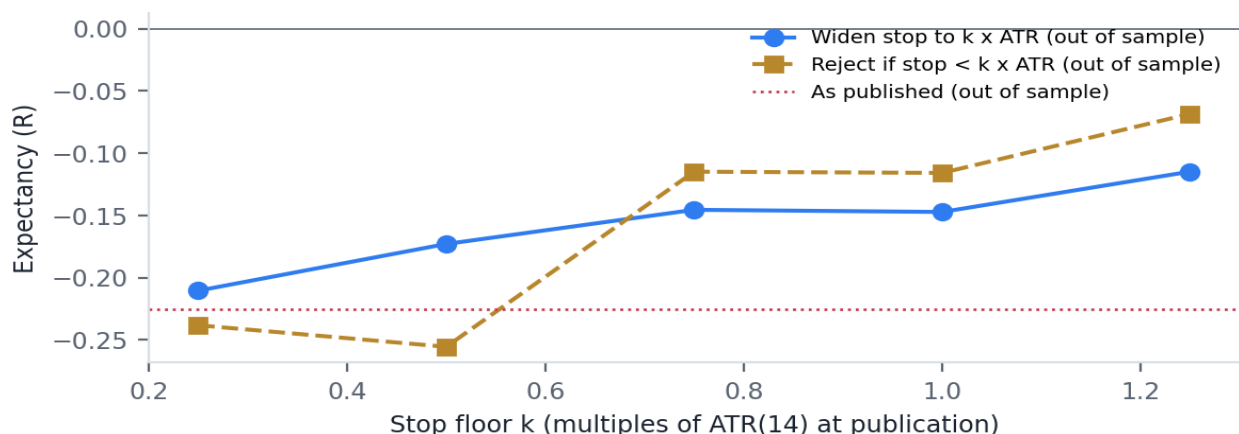


Figure 1. Out-of-sample expectancy by floor. Both interventions beat the published stops at every floor above 0.5x ATR.

Out of sample (2nd half)	Ideas	Expectancy	t	Hit rate	Total R
As published	332	-0.225R	-2.97	30%	-74.8
Widen to 1.25x ATR (shipped)	332	-0.115R	-2.64	41%	-38.1
Reject if under 0.75x ATR	107	-0.115R	-1.15	34%	-12.3

Producer	n	Median stop / ATR	Published	Widened 0.75x	Only stops >= 0.75x (n)
quant	327	0.64	-0.091R	-0.090R	+0.319R (127)
spx_session	179	0.16	-0.268R	+0.024R	+0.000R (0)
gex_scanner	24	0.41	-0.654R	-0.239R	-0.296R (3)
market_scanner	132	1.06	-0.262R	-0.286R	-0.148R (82)

Reading. Widening halves the loss but does not by itself create an edge — selection must still improve. The standout: the quant producer's ideas whose stops were already at least 0.75x ATR averaged **+0.32R** — the only positive slice of the legacy book. The floor is now enforced at the shared ingestion gate for swing and position ideas (day trades exempt: a daily range is the wrong yardstick intraday); every widened idea states the change in its analysis text.

A2. Standing findings carried from v2 (unchanged — no new resolved outcomes since)

Independent 5-minute replay remains the ground truth: legacy book 24.5% hit vs 31.7% chance, -0.191R per idea. SPX-session publishing stays suspended; the tracker's same-day look-ahead bug stays fixed; shorts were not worse than longs once adjudicated. The rebuilt scanners still need ~220 resolved ideas each before any edge claim (power analysis, v2 A5).

Part B — UX 11-7: the SR 11-7 discipline for the interface

B1. The standard

There is no regulator's SR 11-7 for interfaces, so one was assembled from the recognised standards that play the same role: **ISO 9241-11** (usability = effectiveness, efficiency, satisfaction), **ISO 9241-110** (interaction principles), **ISO/IEC 25010** (product quality), **WCAG 2.2 AA** (accessibility), **Core Web Vitals** (speed and stability), **Apple HIG** (touch and safe areas) and **Nielsen's heuristics** (expert review). Each SR 11-7 pillar maps to an interface practice, and each criterion is a pass/fail test run on the live screen by research/ui-scorecard.js.

SR 11-7 pillar		Interface equivalent in UX 11-7	
Model inventory & governance		Screen inventory (13 surfaces x 3 device classes); one design system as the governing policy	
Conceptual soundness		Static audit: tokens not hard-coded colours, type scale, semantic controls (ui-static-audit.py)	
Outcomes analysis		Runtime scorecard on the live site, signed in (ui-scorecard.js)	
Benchmarking		Thresholds from WCAG, Core Web Vitals, HIG — not taste	
Ongoing monitoring		Scorecard re-runnable on any page; results versioned in research/	
Effective challenge / limitations		Findings reported even when unfixed (U8 speed); emulation limits disclosed	
ID	Criterion	Threshold	Standard
U1	Fit	layout width = device width (no zoom-out)	ISO 9241-110 · HIG
U2	Reachable	primary navigation visible on the device	ISO 9241-110 controllability
U3	Legible	>= 95% of text >= 11 px on touch	ISO 9241-303 · HIG
U4	Contrast	>= 98% of text at WCAG AA	WCAG 2.2 1.4.3

ID	Criterion	Threshold	Standard
U5	Targets	>= 90% of controls >= 24 px on touch	WCAG 2.2 2.5.8 · HIG
U6	Named	every control has an accessible name	WCAG 2.2 4.1.2
U7	Stable	CLS <= 0.10	Core Web Vitals
U8	Fast	LCP <= 2.5 s	Core Web Vitals
U9	One system	brand mark present; <= 3 font families	ISO 9241-110 conformity · Nielsen 4
U10	Honest	no 'undefined' / 'NaN' / '[object Object]'	Nielsen 1 · self-descriptiveness
U11	Titled	document title set	WCAG 2.2 2.4.2

B2. iPhone and iPad — what was broken, what changed

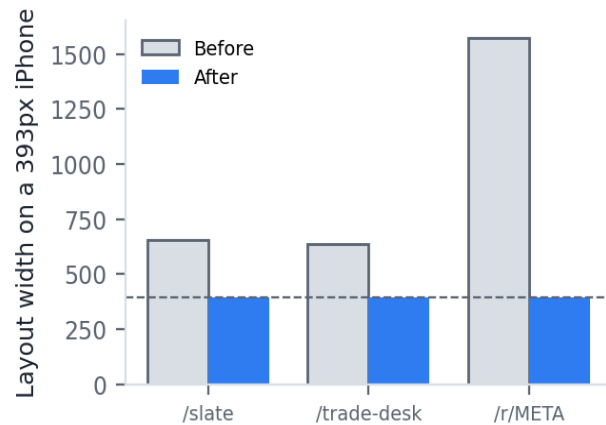


Figure 2. Before, one wide element made iOS zoom the whole page out — dock included — on Slate, Trade Desk and research. Overflow is now contained at the shell, so wide tables scroll inside their own panel.

Finding (iPhone 15 Pro, 393 px)	Fix	Status
Pages zoomed out to 582-1,572 px	Shell contains overflow; frame can't exceed viewport; wrapping tab clusters; fluid Slate grid	Fixed
Research had no bottom dock / no top bar	Research routed through the shared frame	Fixed
778 of 987 board texts under 11 px	Font sizes tokenised (241 CSS + 148 inline); 11 px floor on phones and touch tablets	Fixed
Workup hid evidence + levels rails on phones; header clipped	Rails stack under the chart in one scroll; header wraps	Fixed
Stale workup re-opened on every page, covering the dock	Opens only when a ticker is chosen in the current view	Fixed
Faded muted text 2.9:1	Sub-90% opacity removed from muted text at 453 sites	Fixed
Layout shift: Performance 0.54, Slate 0.29	Viewport-tall placeholder; reserved subtitle height	Fixed
Controls under 24 px on touch	28 px minimum for buttons/tabs under a thumb	Fixed
Third status bar with 'SKYNET v3.2' label	Legacy EngineStatusFooter removed (3 pages)	Fixed

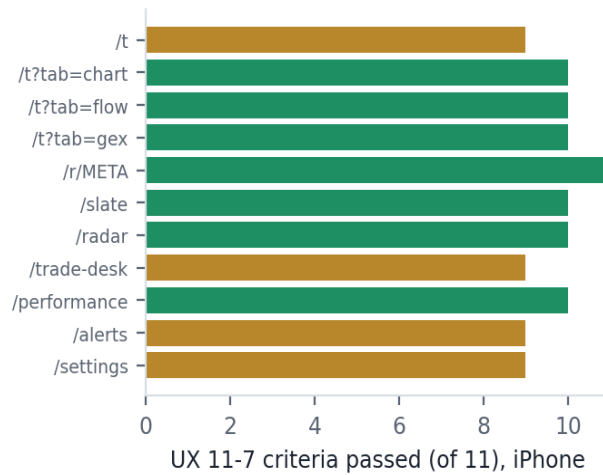


Figure 3. Final scorecard per screen on iPhone. The remaining failure on nearly every data screen is U8 (speed); fit, navigation, legibility, targets, names, honesty and titles pass.

Screen	iPhone	iPad	Remaining failures
/t	9/11	9/11	U8 LCP 6.2s; U9 4 font families
/t?tab=chart	10/11	—	U7 CLS 0.107
/t?tab=flow	10/11	—	U8 LCP 7.0s
/t?tab=gex	10/11	—	U8 LCP 4.0s
/r/META	11/11	10/11	none
/slate	10/11	10/11	U8 LCP 2.8s
/radar	10/11	—	U8 LCP 14.8s
/trade-desk	9/11	7/11	U4 43/44 AA; U8 LCP 5.7s
/performance	10/11	10/11	U8 LCP 2.9s
/alerts	9/11	—	U4 dock labels behind overlay; U8 LCP 6.8s
/settings	9/11	—	U4 disabled button (WCAG-exempt); U8 LCP 2.9s

iPad landscape (1180 px) uses the desktop layout: the unified top navigation fits (14 items in 965 px) with no overflow. The emulator cannot present a touch pointer at that width, so the 11 px floor there was verified by rule (pointer:coarse), not by measurement.

B3. Speed and the server

Measure	Before	After
RAM / free / swap	458 / 8 / 508 MB	1967 / 1024 / 0 MB
Health check	6.43 s	12 ms
/api/market-pulse	21.07 s cold	7 ms (stale-while-revalidate)
/api/extended-hours	8.07 s cold	6 ms
Quote batch (3 symbols) under scanner load	—	5.45 s

Memory is solved. The limit is now the **single vCPU**: load 2.28 with background scanners using ~64% of it, and live quotes queue behind their requests. That is why data-heavy screens still miss the 2.5 s LCP target.

C. Recommendations

#	Action	Why	Owner
1	Move to 2 vCPU (DigitalOcean ~\$18/mo) or cap scanner concurrency	One CPU at load 2.3; quotes wait behind scanners	Operator
2	One aggregated board endpoint instead of ~78 calls	Every call competes for the one CPU; LCP on data screens	Open
3	Weight conviction toward measured-structure evidence	Only structure tags and stops $\geq 0.75 \times$ ATR show positive R	Open
4	Re-run the stop-floor replay monthly as new outcomes resolve	Shipped on 332 out-of-sample ideas; confirm it holds live	Open
5	Migrate remaining raw palette colours to tokens	2,187 raw uses bypass theme and contrast fixes	Open

D. Limitations

Measured overnight with the market closed, in one browser engine, with device emulation rather than physical devices; iPad touch behaviour at landscape width verified by rule. LCP depends on data arrival and varies run to run. The stop-floor test uses one month, one regime, and the same conservative replay rules as v2 (regular session, gaps at the open, double touches as stops). Reproduce: `npx tsx research/stop-floor-test.ts`; inject `research/ui-scorecard.js` on any page and call `__uxScore()`; `python3 research/build-report-v3.py`.